

Level 1 Certification Exam: Prompt and Context Engineering

200 Multiple Choice Questions with Answers

Section A: Fundamentals (Questions 1-30)

1. What is the primary goal of prompt engineering?

- A) Providing relevant documents for grounding
- B) Crafting instructions for desired outputs
- C) Training the AI model
- D) Managing context windows
- **Answer: B**

2. How do Large Language Models (LLMs) fundamentally work?

- A) By understanding human language like humans do
- B) By accessing real-time internet data
- C) As prediction engines guessing next tokens
- D) By storing all possible answers in memory
- **Answer: C**

3. What is context engineering primarily focused on?

- A) Writing better prompts
- B) Providing relevant facts/documents for grounding
- C) Adjusting temperature settings
- D) Creating few-shot examples
- **Answer: B**

4. Which of the following is a common LLM failure mode?

- A) Perfect accuracy
- B) Hallucinations from missing information
- C) Too much creativity

- D) Excessive memory usage
- **Answer: B**

5. What is the recommended workflow when using prompts and context?

- A) Context first, then prompt
- B) Prompt and context simultaneously
- C) Prompt first, then ground with context
- D) Neither matters
- **Answer: C**

6. LLMs are best viewed as:

- A) Human-like intelligence systems
- B) Sophisticated autocomplete systems
- C) Search engines
- D) Databases
- **Answer: B**

7. What causes messy outputs in LLMs?

- A) High temperature settings
- B) Vague instructions
- C) Too much context
- D) Low token limits
- **Answer: B**

8. In agentic flows, what should be done with tool outputs?

- A) Discard them
- B) Feed them back into the system
- C) Save them externally
- D) Ignore them
- **Answer: B**

9. What is the difference between prompt and context engineering?

- A) They are the same thing
- B) Prompts craft instructions, context provides knowledge
- C) Context is only for images
- D) Prompts are deprecated
- **Answer: B**

10. Hallucinations in LLMs typically occur due to:

- A) Too much context
- B) Missing information
- C) High temperature
- D) Long prompts
- **Answer: B**

11. What should you do first when building an AI application?

- A) Add all available context
- B) Set temperature to maximum
- C) Write a clear prompt
- D) Use multiple models
- **Answer: C**

12. Which statement best describes LLM predictions?

- A) They always provide factual information
- B) They guess the most likely next token based on training
- C) They access real-time databases
- D) They understand meaning like humans
- **Answer: B**

13. What type of engineering is more comprehensive and code-like for agents?

- A) Prompt engineering

- B) Software engineering
- C) Context engineering
- D) System engineering
- **Answer: C**

14. Vague instructions lead to:

- A) Better creativity
- B) Unpredictable outputs
- C) Faster responses
- D) Lower costs
- **Answer: B**

15. The primary purpose of grounding with context is to:

- A) Make prompts longer
- B) Reduce hallucinations and provide accurate information
- C) Increase temperature
- D) Slow down responses
- **Answer: B**

16. In prompt engineering, what does "desired output" refer to?

- A) The model's default behavior
- B) The specific result you want from the LLM
- C) The training data
- D) The context window
- **Answer: B**

17. What is a key characteristic of how LLMs process information?

- A) They understand semantics perfectly
- B) They use token prediction based on patterns
- C) They have consciousness

- D) They access external databases automatically
- **Answer: B**

18. When should you provide context to an LLM?

- A) Never
- B) After the prompt fails
- C) When specific knowledge or facts are needed
- D) Only for image generation
- **Answer: C**

19. What happens when an LLM lacks necessary information?

- A) It refuses to answer
- B) It may hallucinate or make up information
- C) It automatically searches the web
- D) It provides a disclaimer only
- **Answer: B**

20. The recommended approach to complex tasks is to:

- A) Use one very long prompt
- B) Break them into chains of simpler steps
- C) Increase temperature significantly
- D) Avoid using any context
- **Answer: B**

21. What makes instructions effective for LLMs?

- A) Being vague and open-ended
- B) Using negative constraints
- C) Being specific and action-oriented
- D) Including unrelated information
- **Answer: C**

22. Prompt engineering is most similar to:

- A) Programming a computer
- B) Having a conversation with instructions
- C) Training a neural network
- D) Managing a database
- **Answer: B**

23. Context engineering is particularly important for:

- A) Simple yes/no questions
- B) Building AI agents and applications
- C) Image generation only
- D) Reducing costs
- **Answer: B**

24. What is the relationship between prompts and tool use?

- A) Prompts replace tools
- B) Tools are called based on prompt instructions
- C) They are unrelated
- D) Tools eliminate the need for prompts
- **Answer: B**

25. The best way to view an LLM's capabilities is:

- A) As unlimited intelligence
- B) As sophisticated pattern completion
- C) As a search engine
- D) As perfect memory
- **Answer: B**

26. What is a recommended first step when outputs are messy?

- A) Add more context

- B) Clarify and specify instructions in the prompt
- C) Lower the temperature
- D) Use a different model
- **Answer: B**

27. Agentic workflows typically involve:

- A) Single-shot prompting only
- B) Iterative steps with tool outputs fed back
- C) No context engineering
- D) Manual intervention only
- **Answer: B**

28. The primary difference between knowledge and understanding in LLMs:

- A) There is no difference
- B) LLMs predict tokens based on patterns, not true understanding
- C) LLMs understand better than humans
- D) Knowledge is impossible for LLMs
- **Answer: B**

29. When crafting prompts, you should focus on:

- A) Making them as long as possible
- B) What output you want to achieve
- C) Using technical jargon
- D) Avoiding examples
- **Answer: B**

30. Context should be provided when:

- A) The LLM's training data is insufficient
- B) You want faster responses
- C) Temperature is high

- D) Using zero-shot prompts
 - **Answer: A**
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Section B: Configuration (Questions 31-50)

31. What does the temperature setting control?

- A) Processing speed
- B) Randomness and creativity in outputs
- C) Context window size
- D) Token cost
- **Answer: B**

32. For consistent, deterministic outputs, you should use:

- A) High temperature
- B) Low temperature (close to 0)
- C) Maximum top-K
- D) No configuration
- **Answer: B**

33. What is top-K sampling?

- A) Selecting the top K documents
- B) Limiting selection to the top K most likely tokens
- C) Using K examples in few-shot
- D) Setting temperature to K
- **Answer: B**

34. Top-P (nucleus sampling) works by:

- A) Selecting tokens until probability threshold is reached
- B) Always picking the top token
- C) Randomly selecting tokens

- D) Limiting output length
- **Answer: A**

35. For mathematical calculations, the recommended temperature is:

- A) 1.0
- B) 2.0
- C) 0 or very low
- D) As high as possible
- **Answer: C**

36. Higher temperature values are appropriate for:

- A) Data analysis
- B) Creative writing
- C) Code generation
- D) Mathematical problems
- **Answer: B**

37. Output/token limits primarily affect:

- A) Accuracy
- B) Length and cost
- C) Temperature
- D) Top-K values
- **Answer: B**

38. A conservative configuration would include:

- A) High temp, high top-P, high top-K
- B) Low temp, low top-P, low top-K
- C) Maximum all settings
- D) Random values
- **Answer: B**

39. What happens when you increase temperature?

- A) Outputs become more deterministic
- B) Outputs become more diverse and random
- C) Processing becomes faster
- D) Cost decreases
- **Answer: B**

40. For code generation, you should typically use:

- A) High temperature for creativity
- B) Low temperature for consistency
- C) Maximum top-P
- D) No configuration limits
- **Answer: B**

41. Top-P value of 0.9 means:

- A) Use 90% of all tokens
- B) Sample from tokens comprising top 90% probability mass
- C) Set temperature to 0.9
- D) Limit output to 90 tokens
- **Answer: B**

42. Balanced configuration typically includes:

- A) Extreme values for all settings
- B) Moderate temperature (~ 0.7), moderate top-P/K
- C) All settings at minimum
- D) Random configuration
- **Answer: B**

43. Token limits are important because they:

- A) Control creativity

- B) Affect cost and response length
- C) Determine accuracy
- D) Set temperature
- **Answer: B**

44. For tasks requiring consistency across multiple runs, you should:

- A) Use high temperature
- B) Use low temperature and same settings
- C) Change settings each time
- D) Ignore configuration
- **Answer: B**

45. What is the relationship between temperature and creativity?

- A) Inverse relationship
- B) Direct relationship (higher temp = more creative)
- C) No relationship
- D) Only affects speed
- **Answer: B**

46. Starting configuration for analytical tasks should be:

- A) Creative (high values)
- B) Conservative (low values)
- C) Balanced
- D) Random
- **Answer: B**

47. Which setting directly controls output length?

- A) Temperature
- B) Top-K
- C) Output/token limits

- D) Top-P
- **Answer: C**

48. For brainstorming sessions, appropriate temperature would be:

- A) 0
- B) 0.2
- C) 0.8-1.0 (higher)
- D) Negative values
- **Answer: C**

49. What is affected by both top-K and top-P?

- A) Output length
- B) Token selection diversity
- C) Processing speed
- D) Cost only
- **Answer: B**

50. Configuration should be chosen based on:

- A) Random preference
- B) The specific task requirements
- C) Model version only
- D) Time of day
- **Answer: B**

Section C: Prompting Techniques (Questions 51-80)

51. Zero-shot prompting means:

- A) Using no prompt
- B) Asking directly without examples
- C) Setting temperature to zero

- D) Using multiple examples
- **Answer: B**

52. How many examples are typically used in few-shot prompting?

- A) 0
- B) 1
- C) 3-5
- D) 100+
- **Answer: C**

53. What is the purpose of few-shot examples?

- A) To increase cost
- B) To show patterns for the LLM to follow
- C) To reduce output length
- D) To set temperature
- **Answer: B**

54. Role prompting involves:

- A) Assigning a persona or expertise to the LLM
- B) Using multiple models
- C) Setting special configuration
- D) Avoiding examples
- **Answer: A**

55. System prompting is used to:

- A) Restart the model
- B) Provide behavior guidelines for the entire conversation
- C) Generate system logs
- D) Configure hardware
- **Answer: B**

56. Chain of Thought (CoT) prompting encourages:

- A) Random responses
- B) Step-by-step reasoning
- C) Shorter answers
- D) No explanation
- **Answer: B**

57. The phrase "Let's think step by step" is used in:

- A) Zero-shot prompting
- B) Chain of Thought prompting
- C) System configuration
- D) Temperature setting
- **Answer: B**

58. Self-Consistency technique involves:

- A) Using the same prompt twice
- B) Generating multiple reasoning paths and picking common answer
- C) Always getting identical outputs
- D) Reducing temperature to zero
- **Answer: B**

59. Step-Back prompting means:

- A) Undoing the previous prompt
- B) Asking a general question before specific ones
- C) Reducing detail
- D) Using fewer examples
- **Answer: B**

60. ReAct prompting combines:

- A) Reasoning and actions/tools

- B) React.js and prompts
- C) Multiple models
- D) Temperature settings
- **Answer: A**

61. Tree of Thoughts (ToT) is useful for:

- A) Simple questions
- B) Complex decisions with branching exploration
- C) Image generation
- D) Setting configuration
- **Answer: B**

62. For consistent reasoning, you should combine CoT with:

- A) High temperature
- B) Temperature = 0
- C) Random sampling
- D) No configuration
- **Answer: B**

63. One-shot prompting uses:

- A) No examples
- B) Exactly one example
- C) Multiple examples
- D) Only questions
- **Answer: B**

64. When would you use role prompting?

- A) For all prompts
- B) When specific expertise or perspective is needed
- C) Never

- D) Only for creative tasks
- **Answer: B**

65. The advantage of Chain of Thought is:

- A) Faster responses
- B) Improved reasoning and transparency
- C) Shorter outputs
- D) Lower cost
- **Answer: B**

66. Self-Consistency improves:

- A) Speed
- B) Accuracy by comparing multiple attempts
- C) Cost efficiency
- D) Output length
- **Answer: B**

67. ReAct is particularly useful for:

- A) Simple Q&A
- B) Tasks requiring tool use and reasoning
- C) Image generation
- D) Setting temperature
- **Answer: B**

68. Tree of Thoughts explores:

- A) Linear paths only
- B) Multiple branching reasoning paths
- C) Single solutions
- D) No reasoning
- **Answer: B**

69. Which technique is best for establishing consistent behavior?

- A) Zero-shot
- B) System prompting
- C) One-shot
- D) High temperature
- **Answer: B**

70. Few-shot examples should be:

- A) Unrelated to the task
- B) Representative of desired patterns
- C) As long as possible
- D) Randomly selected
- **Answer: B**

71. The "step by step" phrase helps with:

- A) Speed
- B) Detailed reasoning
- C) Cost reduction
- D) Image quality
- **Answer: B**

72. When using role prompting, you should:

- A) Be vague about the role
- B) Clearly specify the expertise needed
- C) Change roles mid-conversation
- D) Avoid explaining why
- **Answer: B**

73. Step-Back prompting helps by:

- A) Reducing complexity

- B) Establishing broader context before details
- C) Skipping steps
- D) Using fewer tokens
- **Answer: B**

74. Which technique uses multiple reasoning attempts?

- A) Zero-shot
- B) Self-Consistency
- C) One-shot
- D) System prompting
- **Answer: B**

75. ReAct stands for:

- A) React framework
- B) Reasoning + Acting
- C) Reactive prompting
- D) Real-time actions
- **Answer: B**

76. ToT is most beneficial for:

- A) Simple arithmetic
- B) Complex problem-solving with multiple approaches
- C) Yes/no questions
- D) Image captions
- **Answer: B**

77. System prompts affect:

- A) Only the first response
- B) The entire conversation behavior
- C) Temperature only

- D) Nothing
- **Answer: B**

78. The main benefit of few-shot over zero-shot is:

- A) Lower cost
- B) Better pattern understanding through examples
- C) Faster speed
- D) Shorter prompts
- **Answer: B**

79. CoT with temp=0 provides:

- A) Random reasoning
- B) Consistent step-by-step logic
- C) Creative outputs
- D) No reasoning
- **Answer: B**

80. When would you NOT use Chain of Thought?

- A) For complex reasoning
- B) For simple factual lookups
- C) For math problems
- D) For analysis tasks
- **Answer: B**

Section D: Best Practices & Pitfalls (Questions 81-100)

81. Effective prompts should use:

- A) Vague language
- B) Specific action verbs
- C) Negative instructions only

- D) No structure
- **Answer: B**

82. Positive instructions are better than negative because:

- A) They are shorter
- B) They guide toward desired behavior rather than away from unwanted
- C) They cost less
- D) They are faster
- **Answer: B**

83. Structured formats like JSON are useful for:

- A) Creative writing
- B) Consistent, parseable outputs
- C) Casual conversation
- D) Image generation
- **Answer: B**

84. Using variables in prompts helps with:

- A) Increasing cost
- B) Reusability and templating
- C) Reducing accuracy
- D) Nothing
- **Answer: B**

85. Complex tasks should be:

- A) Handled in one prompt
- B) Broken into chains of simpler steps
- C) Avoided
- D) Done without context
- **Answer: B**

86. Context from prior interactions should be:

- A) Ignored
- B) Provided when relevant for continuity
- C) Always omitted
- D) Randomly included
- **Answer: B**

87. Token optimization is important for:

- A) Accuracy only
- B) Cost and efficiency
- C) Creativity
- D) Speed only
- **Answer: B**

88. Vague or ambiguous instructions lead to:

- A) Better results
- B) Unpredictable outputs
- C) Lower costs
- D) Faster processing
- **Answer: B**

89. Overloading with constraints causes:

- A) Better precision
- B) Confusion and conflicting requirements
- C) Faster outputs
- D) Lower token use
- **Answer: B**

90. Over-relying on tools without reasoning:

- A) Is always best practice

- B) Can miss nuance and context
- C) Reduces costs
- D) Improves accuracy
- **Answer: B**

91. Excessive details or deadlines in prompts:

- A) Always improve quality
- B) Can clutter and confuse the LLM
- C) Are required
- D) Reduce cost
- **Answer: B**

92. A strong prompt for analysis should:

- A) Be vague about what to analyze
- B) Specify data, method, and desired output format
- C) Use only negative constraints
- D) Avoid any structure
- **Answer: B**

93. For coding tasks, effective prompts include:

- A) Vague requirements
- B) Language, requirements, and examples
- C) No examples
- D) Only negative instructions
- **Answer: B**

94. Essay prompts should specify:

- A) Nothing specific
- B) Topic, audience, tone, and format
- C) Only word count

- D) Negative constraints only
- **Answer: B**

95. The best approach to prompt optimization is:

- A) Random changes
- B) Iterative testing and refinement
- C) Using the longest possible prompt
- D) Never changing prompts
- **Answer: B**

96. When instructions conflict:

- A) The LLM always resolves it perfectly
- B) Outputs become unpredictable
- C) Cost increases
- D) Speed improves
- **Answer: B**

97. Structured outputs are achieved by:

- A) Random formatting
- B) Explicitly requesting format (e.g., JSON, tables)
- C) High temperature
- D) Negative instructions
- **Answer: B**

98. Providing context from earlier in conversation:

- A) Always wastes tokens
- B) Helps maintain coherence and build on prior work
- C) Is never useful
- D) Only works with high temperature
- **Answer: B**

99. Negative constraints ("don't do X") should be:

- A) Primarily used
- B) Minimized in favor of positive directions
- C) Always included
- D) Never used
- **Answer: B**

100. A weak prompt typically:

- A) Is specific and clear
 - B) Is vague and lacks direction
 - C) Uses action verbs
 - D) Includes examples
 - **Answer: B**
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Section E: Testing and Evaluation (Questions 101-115)

101. A good testing framework should record:

- A) Nothing
- B) Prompt versions, goals, settings, and quality notes
- C) Only outputs
- D) Only errors
- **Answer: B**

102. A/B testing in prompt engineering means:

- A) Testing two models
- B) Comparing different prompt versions
- C) Using temperature A and B
- D) Testing alphabetically
- **Answer: B**

103. Accuracy as a metric measures:

- A) Speed
- B) Correctness of information
- C) Token count
- D) Creativity
- **Answer: B**

104. Relevance as a metric assesses:

- A) Speed of response
- B) How well output addresses the query
- C) Token cost
- D) Temperature setting
- **Answer: B**

105. Completeness measures whether:

- A) Prompt is long enough
- B) All required aspects are covered in output
- C) Temperature is set
- D) Model is loaded
- **Answer: B**

106. Style consistency is important for:

- A) Single responses only
- B) Maintaining tone across multiple outputs
- C) Temperature only
- D) Cost reduction
- **Answer: B**

107. Format consistency ensures:

- A) All responses are identical

- B) Output structure matches requirements across runs
- C) High creativity
- D) Low cost
- **Answer: B**

108. When evaluating prompts, you should:

- A) Test only once
- B) Test multiple times to assess consistency
- C) Never change them
- D) Ignore outputs
- **Answer: B**

109. Recording quality notes helps with:

- A) Increasing cost
- B) Understanding what works and iterative improvement
- C) Slowing down testing
- D) Nothing
- **Answer: B**

110. Version control for prompts is useful to:

- A) Waste time
- B) Track changes and compare effectiveness
- C) Increase complexity
- D) Reduce accuracy
- **Answer: B**

111. Metrics should be chosen based on:

- A) Random selection
- B) The specific goals of your task
- C) Cost only

- D) Speed only
- **Answer: B**

112. Comparing prompt versions helps:

- A) Increase confusion
- B) Identify most effective approaches
- C) Waste resources
- D) Complicate testing
- **Answer: B**

113. Consistency across runs is tested by:

- A) Running once
- B) Multiple executions with same prompt/settings
- C) Changing settings each time
- D) Using different models
- **Answer: B**

114. Quality notes should include:

- A) Only positive aspects
- B) Both successes and failures with observations
- C) Nothing specific
- D) Only timestamps
- **Answer: B**

115. The goal of evaluation is to:

- A) Prove perfection
- B) Systematically improve prompt effectiveness
- C) Increase costs
- D) Complicate workflows
- **Answer: B**

Section F: Advanced Techniques and Tips (Questions 116-130)

116. In long conversations, context should be:

- A) Always included in full
- B) Summarized to manage token limits
- C) Completely discarded
- D) Randomly selected
- **Answer: B**

117. Prompt chaining involves:

- A) Using one long prompt
- B) Sequential steps where each output feeds the next
- C) Random ordering
- D) Avoiding multiple steps
- **Answer: B**

118. Structured outputs like JSON are valuable for:

- A) Creative writing
- B) Analysis tasks requiring parsing and processing
- C) Casual chat
- D) Image generation
- **Answer: B**

119. In multi-modal prompting with images, you should:

- A) Assume LLM sees everything
- B) Be explicit about what to look for in images
- C) Never describe images
- D) Use only text
- **Answer: B**

120. When context becomes too long:

- A) Include everything anyway
- B) Summarize or break into focused chunks
- C) Start over completely
- D) Ignore context
- **Answer: B**

121. Breaking complex tasks into steps:

- A) Wastes time
- B) Improves accuracy and manageability
- C) Always fails
- D) Increases cost only
- **Answer: B**

122. Output from one prompt can be:

- A) Never reused
- B) Input for subsequent prompts in a chain
- C) Only stored externally
- D) Discarded immediately
- **Answer: B**

123. Structured output formats help with:

- A) Creative expression
- B) Programmatic processing of results
- C) Reducing accuracy
- D) Nothing useful
- **Answer: B**

124. For image analysis, prompts should:

- A) Be vague

- B) Specify what elements to focus on
- C) Never mention the image
- D) Use maximum temperature
- **Answer: B**

125. Context management techniques include:

- A) Including everything always
- B) Summarization, chunking, and prioritization
- C) Deleting all history
- D) Random selection
- **Answer: B**

126. Sequential prompting is useful when:

- A) Task is simple
- B) Task requires multiple distinct steps
- C) No context is needed
- D) Time is unlimited
- **Answer: B**

127. Explicit instructions in multi-modal prompts:

- A) Are unnecessary
- B) Help focus analysis on relevant details
- C) Reduce accuracy
- D) Only increase cost
- **Answer: B**

128. JSON outputs are particularly useful for:

- A) Poetry
- B) Data that will be processed programmatically
- C) Casual conversation

- D) Storytelling
- **Answer: B**

129. Summarizing past conversation helps:

- A) Lose information
- B) Retain key points while managing tokens
- C) Confuse the LLM
- D) Increase costs
- **Answer: B**

130. The benefit of prompt chains is:

- A) Simplicity only
- B) Breaking complexity into manageable, logical steps
- C) Random outputs
- D) Avoiding structure
- **Answer: B**

Section G: Mixture-of-Experts (MoE) & Prompting (Questions 131-145)

131. In MoE architecture, what selects which experts to use?

- A) Random selection
- B) Gating network/router
- C) User manually
- D) Temperature setting
- **Answer: B**

132. What does "sparse activation" in MoE mean?

- A) All experts always active
- B) Only subset of experts activated per input
- C) No experts used

- D) Random activation
- **Answer: B**

133. MoE benefits include:

- A) Simplicity only
- B) Efficiency and specialization
- C) Always slower
- D) Higher costs always
- **Answer: B**

134. A drawback of MoE is:

- A) Too simple
- B) Routing instability and memory overhead
- C) Perfect accuracy
- D) No specialization
- **Answer: B**

135. For MoE models, domain-specific signals should be:

- A) Hidden
- B) Provided upfront in prompt
- C) Omitted
- D) Placed randomly
- **Answer: B**

136. When using MoE, mixed tasks should be:

- A) Combined always
- B) Separated for better routing
- C) Avoided completely
- D) Randomly mixed
- **Answer: B**

137. Front-loading cues in MoE prompts helps:

- A) Nothing
- B) Router select appropriate experts early
- C) Confuse routing
- D) Increase cost
- **Answer: B**

138. For consistency in MoE models, you should:

- A) Use high temperature
- B) Reduce temperature
- C) Use maximum variance
- D) Ignore configuration
- **Answer: B**

139. Expert-aware prompting means:

- A) Random phrasing
- B) Using explicit language/style signals
- C) Avoiding clarity
- D) Generic instructions
- **Answer: B**

140. Vagueness in MoE prompts causes:

- A) Better routing
- B) Poor expert selection
- C) Optimal performance
- D) Lower cost
- **Answer: B**

141. MoE architectures use sub-networks called:

- A) Layers

- B) Experts
- C) Routers only
- D) Prompts
- **Answer: B**

142. Routing instability in MoE refers to:

- A) Network errors
- B) Inconsistent expert selection for similar inputs
- C) Perfect routing
- D) Fast processing
- **Answer: B**

143. Memory overhead in MoE is due to:

- A) Single expert
- B) Multiple experts stored in memory
- C) Small models
- D) No parameters
- **Answer: B**

144. To prompt MoE effectively, you should:

- A) Be vague
- B) Front-load domain signals and be explicit
- C) Hide intent
- D) Use random language
- **Answer: B**

145. Sparse activation provides:

- A) Slower performance
- B) Efficiency by using only needed experts
- C) All experts always

- D) No specialization
 - **Answer: B**
-

Section H: Practical Application (Questions 146-160)

146. For content creation, effective prompts should specify:

- A) Nothing
- B) Topic, audience, tone, and format
- C) Only word count
- D) Temperature only
- **Answer: B**

147. Code generation prompts should include:

- A) Vague requirements
- B) Language, requirements, and examples
- C) Only the language
- D) No details
- **Answer: B**

148. For customer feedback analysis, prompts should ask for:

- A) Nothing specific
- B) Sentiment, themes, and recommendations
- C) Only positive feedback
- D) Random insights
- **Answer: B**

149. When creating content, audience specification helps:

- A) Nothing
- B) Tailor tone and complexity appropriately
- C) Increase confusion

- D) Reduce quality
- **Answer: B**

150. Code requirements should be:

- A) Vague and open-ended
- B) Specific about functionality, inputs, and outputs
- C) Omitted entirely
- D) Only about language choice
- **Answer: B**

151. Tone specification in content prompts:

- A) Is unnecessary
- B) Guides the style and voice of output
- C) Only affects length
- D) Reduces accuracy
- **Answer: B**

152. For analysis tasks, you should specify:

- A) Nothing
- B) What to analyze, method, and output format
- C) Only the data
- D) Temperature only
- **Answer: B**

153. Examples in code prompts help:

- A) Confuse the LLM
- B) Clarify expected patterns and style
- C) Increase errors
- D) Waste tokens
- **Answer: B**

154. Customer feedback prompts should:

- A) Focus on complaints only
- B) Request balanced analysis of themes and sentiment
- C) Ignore patterns
- D) Be one word long
- **Answer: B**

155. Format specification is crucial for:

- A) Casual chat
- B) Outputs that need specific structure (reports, JSON, tables)
- C) Creative fiction only
- D) Nothing
- **Answer: B**

156. When generating content, providing context about purpose:

- A) Wastes tokens
- B) Helps align output with intended use
- C) Reduces quality
- D) Is never needed
- **Answer: B**

157. For code debugging prompts, include:

- A) Nothing about the error
- B) Error message, code snippet, and expected behavior
- C) Only the code
- D) Random information
- **Answer: B**

158. Recommendations from analysis should be:

- A) Vague

- B) Actionable and specific
- C) Omitted
- D) Random
- **Answer: B**

159. Multi-step content creation benefits from:

- A) Single massive prompt
- B) Breaking into outline, drafting, and refinement steps
- C) No planning
- D) Random approach
- **Answer: B**

160. For technical documentation, specify:

- A) Nothing
- B) Audience expertise level, format, and coverage
- C) Only topic
- D) Random details
- **Answer: B**

Section I: Context Engineering Integration (Questions 161-175)

161. RAG stands for:

- A) Random Access Generation
- B) Retrieval-Augmented Generation
- C) Rapid AI Gateway
- D) Router Agent Generation
- **Answer: B**

162. In RAG, you should prioritize:

- A) Oldest passages

- B) Relevant and newest passages
- C) Random passages
- D) All passages equally
- **Answer: B**

163. Chunking in RAG refers to:

- A) Random splitting
- B) Breaking documents into optimal-sized segments
- C) Deleting content
- D) Combining everything
- **Answer: B**

164. Ranking in RAG helps:

- A) Slow down processing
- B) Order passages by relevance
- C) Randomize results
- D) Delete content
- **Answer: B**

165. Deduplication in RAG is important to:

- A) Increase token use
- B) Avoid redundant information
- C) Complicate processing
- D) Reduce accuracy
- **Answer: B**

166. Token budgets in RAG manage:

- A) Financial costs only
- B) How much context can be included
- C) Nothing

- D) Temperature
- **Answer: B**

167. When combining prompts and context:

- A) Use only prompts
- B) Prompts define behavior, context provides knowledge
- C) Use only context
- D) They can't be combined
- **Answer: B**

168. Optimal chunk size in RAG balances:

- A) Nothing
- B) Context completeness and token efficiency
- C) Speed only
- D) Random factors
- **Answer: B**

169. RAG retrieval should focus on:

- A) All available documents
- B) Most semantically relevant passages
- C) Random selection
- D) Oldest documents
- **Answer: B**

170. Context for RAG should be:

- A) Unfiltered and complete
- B) Filtered and ranked by relevance
- C) Random
- D) Minimal always
- **Answer: B**

171. Combining RAG with prompts improves:

- A) Nothing
- B) Accuracy by grounding in specific knowledge
- C) Only speed
- D) Only creativity
- **Answer: B**

172. Token budget optimization involves:

- A) Using unlimited tokens
- B) Prioritizing most relevant content within limits
- C) Random selection
- D) Ignoring limits
- **Answer: B**

173. In RAG systems, prompts should:

- A) Ignore the retrieved context
- B) Instruct how to use the retrieved context
- C) Replace context
- D) Be omitted
- **Answer: B**

174. Semantic search in RAG finds:

- A) Exact keyword matches only
- B) Meaning-based relevant passages
- C) Random documents
- D) Nothing useful
- **Answer: B**

175. RAG is most useful when:

- A) No specific knowledge needed

- B) Task requires grounding in specific documents/data
 - C) Creating fiction
 - D) Simple arithmetic
 - **Answer: B**
-

Section J: Context Engineering Deep Dive (Questions 176-190)

176. In the LLM-as-CPU analogy, the context window is like:

- A) Hard drive
- B) RAM (working memory)
- C) CPU cores
- D) Network connection
- **Answer: B**

177. Context engineering is particularly important for:

- A) Simple queries
- B) Building AI agents and applications
- C) Single-word answers
- D) Nothing
- **Answer: B**

178. The difference between context and prompt engineering is:

- A) No difference
- B) Context is more comprehensive and code-like for agents
- C) Prompts are always better
- D) Context is deprecated
- **Answer: B**

179. AI agent components include:

- A) Only the model

- B) Model, tools, memory, knowledge, orchestration, guardrails
- C) Just prompts
- D) Nothing specific
- **Answer: B**

180. Tools in agents allow:

- A) Nothing
- B) External interaction (APIs, searches, etc.)
- C) Only internal processing
- D) Temperature changes
- **Answer: B**

181. Memory in agents stores:

- A) Nothing
- B) Dynamic conversation history
- C) Only the model
- D) Random data
- **Answer: B**

182. Knowledge in agents represents:

- A) Dynamic memory
- B) Static information and facts
- C) Tools
- D) Prompts only
- **Answer: B**

183. Guardrails in agents ensure:

- A) Nothing
- B) Behavior stays within safety and policy boundaries
- C) Maximum creativity

- D) Random outputs
- **Answer: B**

184. Multi-agent systems work by:

- A) Using one agent only
- B) Splitting tasks across specialized agents
- C) Random distribution
- D) Avoiding coordination
- **Answer: B**

185. Context isolation is important for:

- A) Mixing everything
- B) Security and preventing information leakage
- C) Slowing down
- D) Nothing
- **Answer: B**

186. Hierarchical summarization helps with:

- A) Nothing
- B) Compressing long contexts while retaining key info
- C) Expanding content
- D) Random processing
- **Answer: B**

187. State management in agents tracks:

- A) Nothing
- B) Current conversation state and variables
- C) Only prompts
- D) Random values
- **Answer: B**

188. Semantic chunking in RAG:

- A) Randomly splits text
- B) Divides based on meaning and coherence
- C) Uses fixed character counts only
- D) Avoids splitting
- **Answer: B**

189. Temporal reasoning in agents involves:

- A) Ignoring time
- B) Understanding sequence and time-based relationships
- C) Random ordering
- D) No logic
- **Answer: B**

190. Enterprise context engineering requires:

- A) Simple approaches only
 - B) Modular components balancing accuracy, latency, cost
 - C) No planning
 - D) Random architecture
 - **Answer: B**
-

Section K: Six-Step Framework (Questions 191-200)

191. The first step in the 6-step framework uses:

- A) Weak verbs
- B) Strong command verbs (analyze, create, etc.)
- C) No verbs
- D) Random words
- **Answer: B**

192. The "Rule of Three" for context asks:

- A) Three random questions
- B) Who/what/when or past/present/future
- C) Three temperatures
- D) Three models
- **Answer: B**

193. The Logic step specifies:

- A) Nothing
- B) Reasoning approach and output format
- C) Only temperature
- D) Random methods
- **Answer: B**

194. Roleplay in the framework means:

- A) Acting randomly
- B) Specifying the expertise or perspective needed
- C) Avoiding expertise
- D) Using multiple models
- **Answer: B**

195. The Questions step involves:

- A) Avoiding questions
- B) Iterative refinement until repetition indicates clarity
- C) One question only
- D) Random queries
- **Answer: B**

196. Voice memos in the framework allow:

- A) Nothing

- B) Natural, conversational follow-ups
- C) Only text
- D) Random audio
- **Answer: B**

197. Extensive context should be used when:

- A) Task is simple
- B) Task is complex and needs background
- C) Always
- D) Never
- **Answer: B**

198. Minimal context is appropriate for:

- A) Complex analysis
- B) Simple, straightforward tasks
- C) All tasks
- D) Nothing
- **Answer: B**

199. Great prompters differ from weak ones by:

- A) Using longer prompts
- B) Using questions to iteratively refine
- C) Avoiding clarification
- D) Random approaches
- **Answer: B**

200. Common mistakes in the framework include:

- A) Clear commands
- B) Vague commands and generic information
- C) Strong verbs

- D) Specific context
 - **Answer: B**
-

BONUS: Image Generation (Nano Banana Principles)

201. In image generation prompts, specificity is:

- A) Discouraged
- B) Preferred over generality
- C) Irrelevant
- D) Random
- **Answer: B**

202. Visual hierarchy in image prompts starts with:

- A) Technical details
- B) Subject description
- C) Mood
- D) Random elements
- **Answer: B**

203. Professional photography language includes:

- A) Vague descriptions
- B) Camera terminology and lighting setups
- C) No technical terms
- D) Random words
- **Answer: B**

204. An 85mm lens is typically used for:

- A) Wide landscapes
- B) Portraits with pleasing compression
- C) Macro photography

- D) Random shots
- **Answer: B**

205. Aperture f/1.4 creates:

- A) Everything in focus
- B) Shallow depth of field
- C) No effect
- D) Random results
- **Answer: B**

206. "Of the uploaded photo" helps with:

- A) Nothing
- B) Consistency and anchoring to source
- C) Random variation
- D) Slowing down
- **Answer: B**

207. Lighting patterns like "Rembrandt" refer to:

- A) Random lighting
- B) Specific professional lighting setups
- C) No lighting
- D) Natural only
- **Answer: B**

208. Corporate/professional image style includes:

- A) Casual snapshots
- B) Executive portraits and headshots
- C) Abstract art
- D) Random styles
- **Answer: B**

209. Shallow depth of field is used to:

- A) Show everything
- B) Isolate subject from background
- C) Blur everything
- D) Random effect
- **Answer: B**

210. Common image prompt mistakes include:

- A) Clear specifications
- B) Being too vague
- C) Professional language
- D) Specific lighting
- **Answer: B**